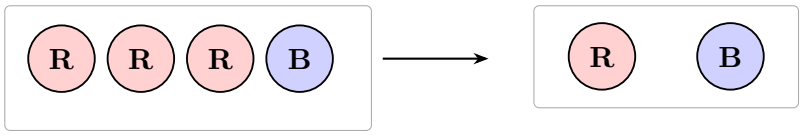
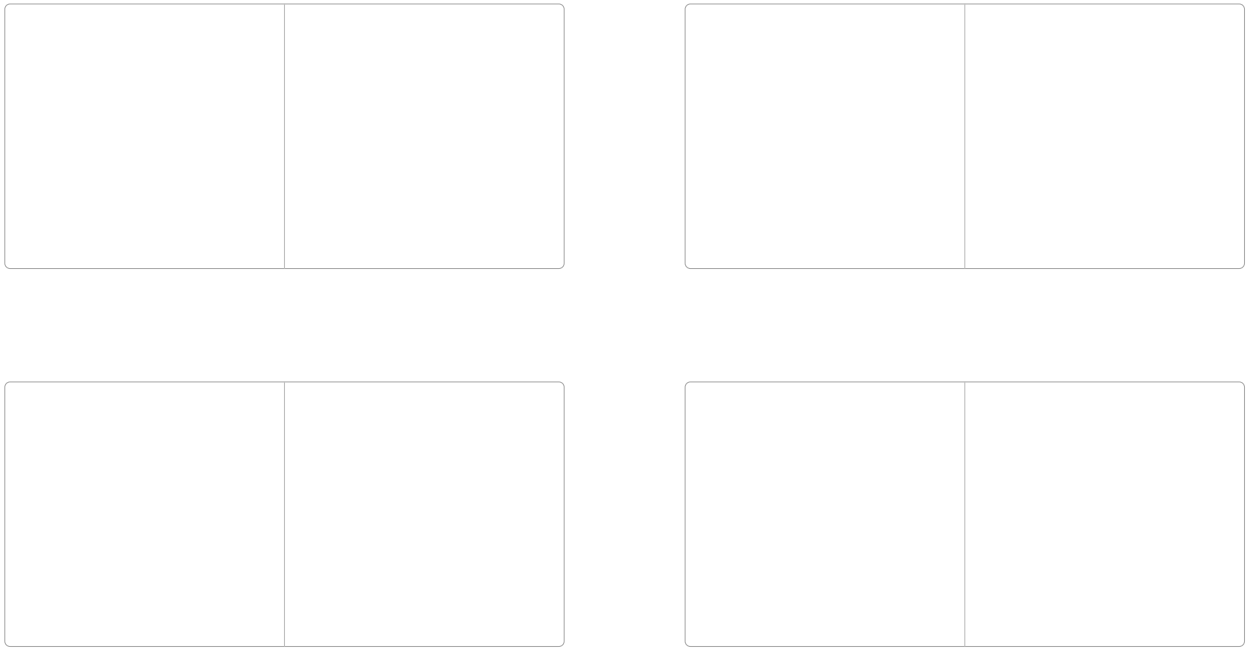
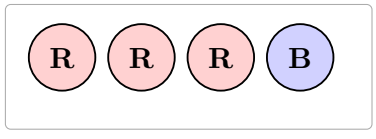
Use the same bag. Put each counter back after every draw; for random draws, mix and draw without looking.



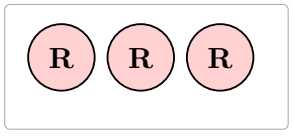
Red, then blue makes this pair. Keep the order.

Problem 1: Choose draws yourself to make every different color pair. Which pairs disappear if you take the blue counter out of the bag?



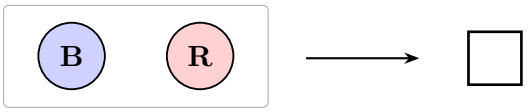


the bag



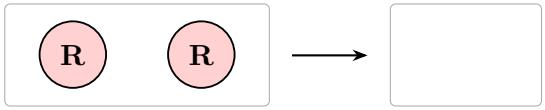
blue taken out

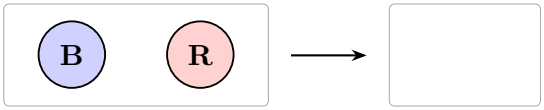
Use three red counters and one blue again. A rule gives each color pair a square, a circle, or no shape. The same color pair always gets the same answer.

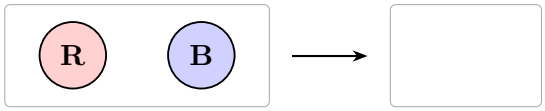


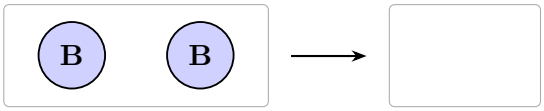
A sample rule: BR gives a square.

Problem 2: Make a rule that sometimes gives each shape and gives them the same chance. Try it on six random pairs.





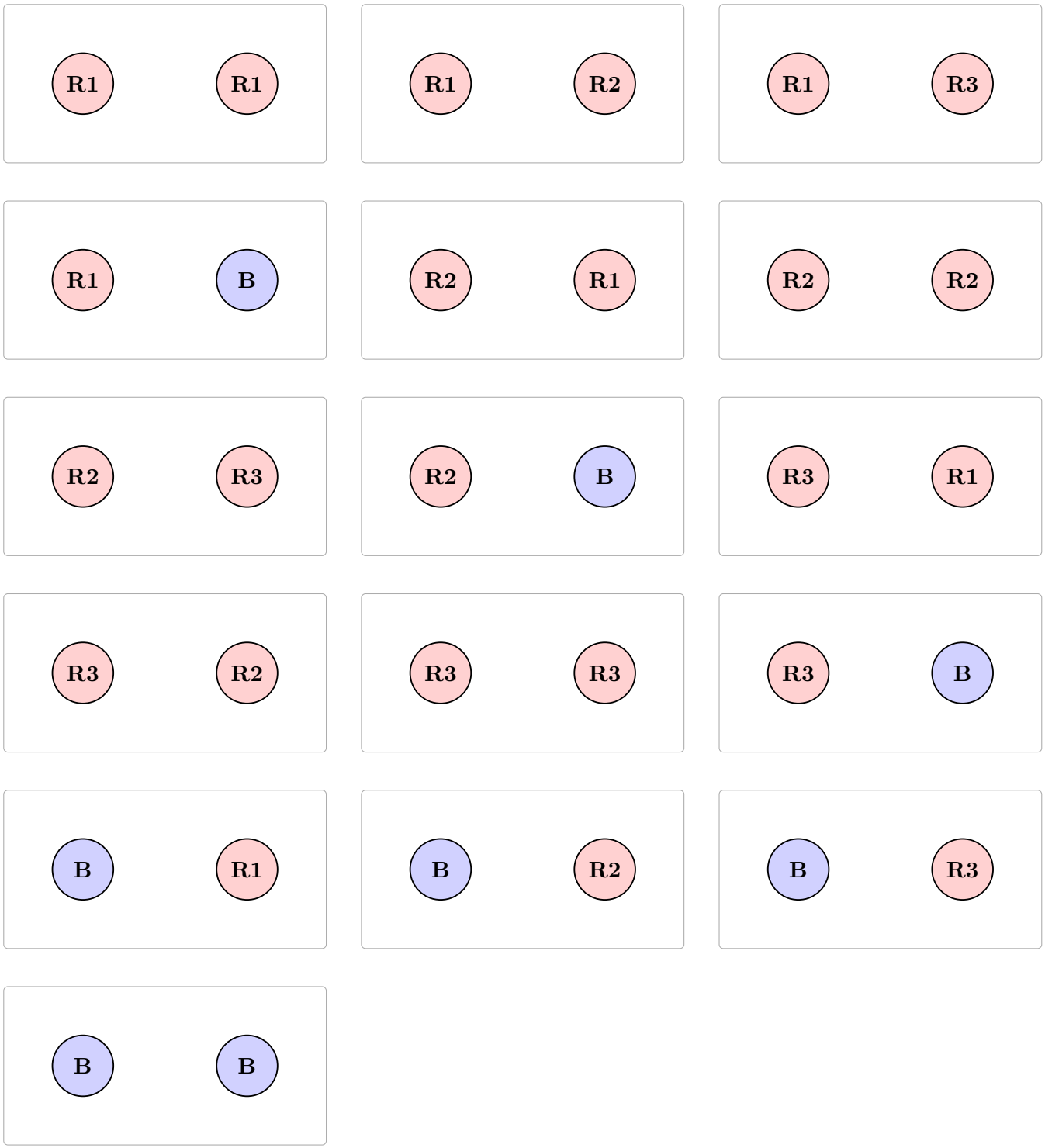




Use a cross to record no shape.

The red counters have marks R1, R2, R3. Each marked pair below has the same chance.

Problem 3: Does your rule give each shape to the same number of pictures? Find every rule that does and sometimes gives each shape.



Problem 4: Keep your rule for these bags. Which can give each shape, and do the shapes have the same chance?



Marked pairs from R, B1, B2, B3

R R	R B1	R B2	R B3
B1 R	B1 B1	B1 B2	B1 B3
B2 R	B2 B1	B2 B2	B2 B3
B3 R	B3 B1	B3 B2	B3 B3

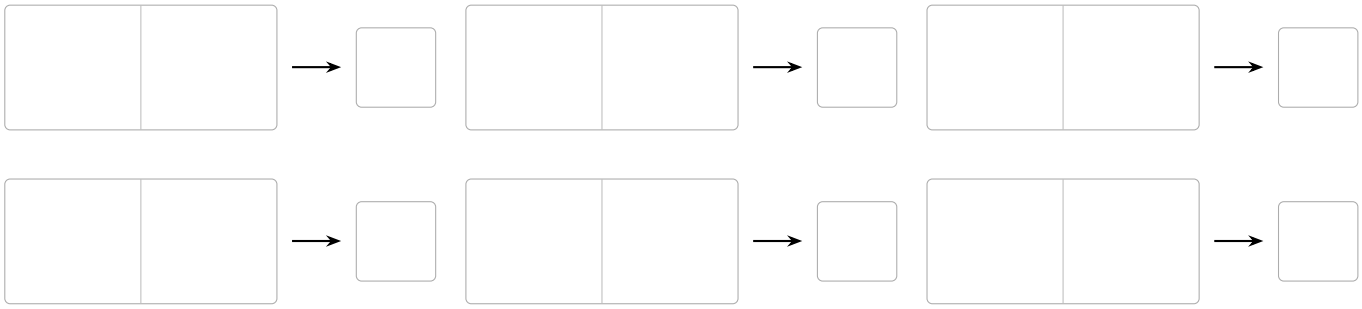
Marked pairs from R1, R2, B1, B2

R1 R1	R1 R2	R1 B1	R1 B2
R2 R1	R2 R2	R2 B1	R2 B2
B1 R1	B1 R2	B1 B1	B1 B2
B2 R1	B2 R2	B2 B1	B2 B2

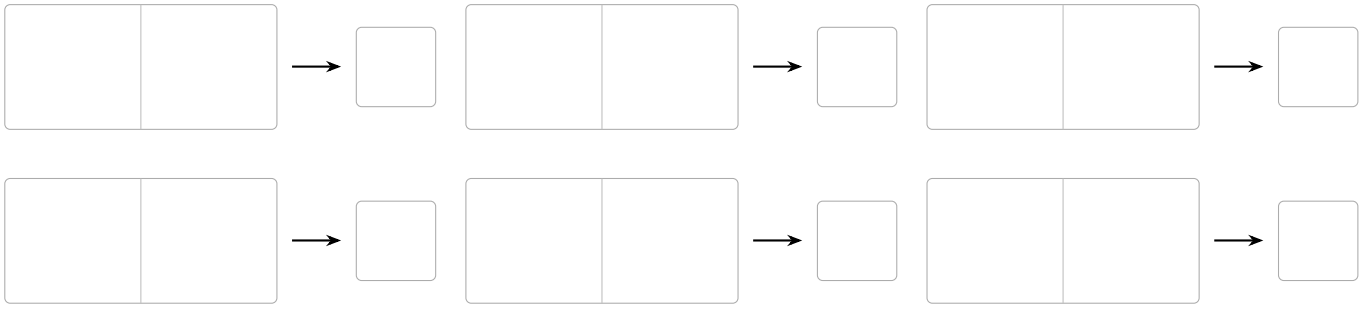
Problem 5: Keep your rule. Choose a four-counter bag that makes shapes come out as often as possible. Could any four-counter bag give a shape on every pair?

Problem 6: Choose pairs and draw six-pair stories using your rule for each target. Do six random pairs always give a shape?

no shapes



exactly one square and no circles



exactly three circles and no squares

